

Machine Learning and Predictive Analytics

Couework Mark Sheet



STUDENT:

Summary

Criteria	Mark (out of 10 per section)
Executive Summary (10%)	
Initial analysis of churn rates and provision of churn definition (5%)	
Feature selection, prediction approach and evaluation (50%)	
Insight report part 1 (10%)	
Insight report part 2 (25%)	

OVERALL SCORE (out of 100) (rounding applied)	
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Criteria 1: Executive Summary (10%)

0-4.9	5-5.9	6-6.5	6.5-7	7-8	8-9	9-10
Incoherent summarization which does not convey what is contained in the report.	Limited summarisation of the analysis. Key findings stated, but it is unclear why the findings were reached.	Summarization of finding at an appropriate level though could be improved. Somewhat clear what analytics was undertaken to reach the conclusions.	Summarization of findings was at an appropriate level. Generally clear what analytics was undertaken to reach the conclusions.	Clear summary of findings. It is clear what analytics were used to reach the conclusions.	Findings fully summarized in a clear and succinct manner. The analytics process to reach these was clear and succinctly described.	Professional level executive summary.

Comments:

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Mark	

Criteria 2: Churn Rate (5%)

0-4.9	5-5.9	6-6.5	6.5-7	7-8	8-9	9-10
Justification of why the selected churn value is omitted or flawed. Incorrect interpretation of provided report.	Justification of why the selected churn value is present but unclear and/or contains minor errors. Correct interpretation of the provided report with potentially some errors/misunderstandings.	Relatively clear justification of the selected churn definition. Mostly sound interpretation of the provided report. Some aspects could have been improved to make the report more concise.	Clear justification of the selected churn definition. Sound interpretation of the provided report. Some aspects could have been improved to make the report more concise.	Clear justification of the selected churn definition. Clear and fully correct interpretation of the provided report. Some minor aspects could have been improved to make the report more concise.	Succinct justification of the selected churn definition with full justification. Justification clearly and concisely linked to the outcomes from the report and the problem.	Industry/consultant quality work fully articulating and justifying the choice of churn rate with regard to the report and problem domain.

Comments:

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Mark	

Criteria 3: Feature selection, prediction approach and evaluation (50%). Feature engineering must be predominantly in SQL, rest in Python.

0-4.9	5-5.9	6-6.5	6.5-7	7-8	8-9	9-10
The report and analysis does not show an understanding of the key concepts. Major errors in understanding and in implementation.	An effort at an analysis that shows understanding of key concepts. However, the report is poorly written, lacks vital information and/or has a number of minor errors. The evaluation method use out-of-time sampling. Potentially some procedural overfitting.	A good exploration or justification of some, but not all, aspects of the machine learning pipeline. This includes feature choice & selection, preprocessing and model selection. Use of out-of-time sampling for testing. No procedural overfitting.	A good exploration or justification of almost all aspects of the machine learning pipeline. This includes feature choice & selection, preprocessing and model selection. Correct use of out-of-time sampling for testing. No procedural overfitting.	A good range of temporal and non-temporal features implemented and justified. Feature selection considered with no errors. A good range of model variants considered and justified. Meta-parameters tuned with a validation set. Out-of-time evaluation correctly used. No procedural overfitting.	A good range of temporal and non-temporal features implemented and justified. Feature selection considered with no errors. A good range of model variants considered and justified. Meta-parameters tuned with a validation set. Repeated out-of-time evaluation correctly used. No procedural overfitting.	Industry/consultant quality work, both with regard to the report and code quality.

Comments:

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Mark	

Criteria 4: Insight Report Part 1 (10%)

0-4.9	5-5.9	6-6.5	6.5-7	7-8	8-9	9-10
The summary fails to present the analysis to a marketer in any meaningful way.	Most points are summarized. but retains too much technical language.	Key points are summarized. in language / structure mostly appropriate for a marketing department. Some depth regarding the findings missing.	Key points are summarized. in language / structure appropriate for a marketing department. Some depth regarding the findings missing.	All applicable points are summarized in language / structure appropriate for a marketing department.	All applicable points are summarized succinctly in language / structure appropriate for a marketing department.	Industry/consultant quality summarization of findings. Clear and concise with language / structure appropriate for a marketing department.

Comments:

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Mark	

Criteria 5: Insight Report Part 2 (25%)

0-4.9	5-5.9	6-6.5	6.5-7	7-8	8-9	9-10
Limited and/or incorrect analysis.	Simple summary statistics used to describe churner vs. non-churners with no errors. Alternatively more in-depth summary statistics were attempted with minor errors and/or issues with their description.	Variable importance used to focus discussion of summary statistics and generate pen portraits. Minor errors or poor choice of variable importance measure.	Variable importance used to focus discussion of summary statistics and generate pen portraits.	Variable importance used to focus discussion of summary statistics and generate pen portraits. Reflection on the findings with respect to feature selection / the ML pipeline and interpretation for insights regarding churn and non-churn customers.	In depth analysis of churner vs. non-churns beyond variable importance and summary statistics drawing relevant conclusions.	Industry/consultant quality work, both with regard to the report and code quality.

Comments:

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Mark	